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Set Operations Documentation

I created a Set Operations class that has methods for performing set operations on Integer ArrayLists, as well as calculating probabilities related to independent and dependent events. To test these methods I used an Integer ArrayList for a set and another for a subset to pass through the set operation methods:

```
ArrayList<Integer> set = new ArrayList<>();  
set.add(1);  
set.add(2);  
set.add(3);  
set.add(4);  
set.add(5);  
set.add(6);  
set.add(7);  
set.add(8);  
set.add(9);  
set.add(10);  
set.add(11);  
set.add(12);  
set.add(13);
```

```
ArrayList<Integer> subset = new ArrayList<>();  
subset.add(2);  
subset.add(4);  
subset.add(6);  
subset.add(8);
```

When unioning these sets I got:

```
Union: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13]
```

For intersection I got:

```
Intersect: [2, 4, 6, 8]
```

The complement of the subset is:

```
Complement: [1, 3, 5, 7, 9, 10, 11, 12, 13]
```

These were the specific probabilities passed into the independent and dependent intersection methods to find the probabilities of independent and dependent intersection

```
independentIntersection( probA: .43, probB: .57));  
  
dependentIntersection( probA: .43, conditionalProbBGivenA: .57));
```

The probability of independent intersection:

```
Independent Intersection: 0.24509999999999998
```

The probability of dependent intersection:

```
Dependent Intersection: 0.24509999999999998
```

These were the specific probabilities passed into the exclusive and nonexclusive union methods to find the probabilities of exclusive and nonexclusive union

```
+ exclusiveUnion( probA: .66, probB: .34));  
  
" + nonExclusiveUnion( probA: .66, probB: .34, intersectionProb: .22));
```

The probability of exclusive union:

```
Exclusive Union: 1.0
```

The probability of nonexclusive union:

```
NonExclusive Union: 0.78
```